Homework 1

Solve the following problems justifying all your answers. You should not use a graph calculator to justify any answer.

1. Solve the following:

(a) $|\frac{2x-1}{x+1}| = 3$,

(b) $|x+2| + |x-3| \leq 10$

2. Sketch the graph of the following function:

$$f(x) = \begin{cases} \frac{-|1-4x|}{2} & \text{if} \quad x < 0, \\ \sqrt{1 - \frac{(x-3)^2}{9}} - \frac{1}{2} & \text{if} \quad 0 \leq x < 6, \\ \sqrt{1 + (x-6)^2} - \frac{1}{2} & \textbf{if} \ \ x \geq 6 \end{cases} \tag{1}$$

3. Find the maximal possible domain and the image of the following functions:

(a) $f(x) = \frac{4x}{x^2+4}$,

(b) $g(x) = 4 - \sqrt{x^2 + 4x - 5}$

4. Sketch the following regions in the plane:

(a) $\left\{(x,y) \in \mathbb{R}^2 : (x-3)^2 + (y-2)^2 \leq 4,\ y < 2x - 6,\ y \geq -\frac{3}{4}x + 3\right\}$,

(b) $\left\{(x,y) \in \mathbb{R}^2 : |x-2| + |y-3| \leq 10\right\}$